

Algorithmic Trading SoS 2025

2-Month Weekly Learning Strategy

Week 1: Introduction to Investing & Stock Markets

- Read: 'The Need to Invest' - Zerodha Varsity
- Research: What are Stocks? What is the Stock Market?
- Watch: Market Participants - ICICI Direct

Week 2: Candlesticks, Volumes & Price Movements

- Read: Candlestick Charts (CashOverflow), Volume (5Paisa)
- Watch: Order Book (YouTube), Bid-Ask Spread (Investopedia), Stock Market Index (YouTube)

Week 3: Fundamentals & Financial Ratios

- Read: Financial Ratios (CFI)
- Watch: Fundamental Analysis, Financial Ratios (YouTube)

Week 4: Reading Financial Statements

- Watch: Annual Report, Cash Flow, Income Statement, Balance Sheet, Connecting Statements

Week 5: Technical Analysis Overview

- Read: Types of Trading Indicators - QuantShare
- Watch: Technical Indicators - YouTube

Week 6: Deep Dive into Algo Trading & Order Types

- Read: Introduction to Algo Trading, Order Types
- Watch: Zerodha Varsity videos/blogs, CA Rachna Ranade on TradingView, Trading Perspectives video

Week 7: Strategy Design + Python Basics

- Learn: Momentum, Trend, Volume, Volatility indicators
- Read/Watch: Combining Technical Indicators
- Python: Basics of Numpy, Pandas, Matplotlib

Week 8: Backtesting and Real Implementation

- Tools: Backtrader (Docs, Articles), TradingView (Pine Script), Zerodha Streak
- Practice: Jupyter notebook from Backtrader, TradingView strategies